Name: Krish Idnani Batch: T12 Roll No. 42

Experiment 10

Aim: Learn Dockerfile Instruction and Build Image

Step 1: Downloading Docker

The first place to start is the **official Docker website** from where we can download Docker Desktop.

Step 2: Configuration

To run Linux on Windows, Docker requires a virtualization engine. Docker recommends using WSL 2

Step 3: Running the installation.

Click Ok, and wait a bit…

Step 4: Restart

For Docker to be able to properly register with Windows, a restart is required at this point.

Step 5: WSL 2 installation

After you accept the license terms, the Docker Desktop window will open. However, we are not done yet. Since we have selected WSL 2 as our virtualization engine, we also need to install it. **Don't click Restart just yet!**

Step 6— Starting Docker Desktop

If Docker Desktop did not start on its own, simply open it from the shortcut on your Desktop. If you wish, you can do the initial orientation by clicking Start.

Step 7— Testing Docker

Open your favourite command line tool and type in the following command: docker run hello-world.

This will download the hello-world Docker image and run it. This is just a quick test to ensure everything is working fine.

Step 8 — Automatically start Docker [Optional]

This step is dependently optional, but if you are working a lot with Docker, you may want to configure Docker Desktop to start automatically once you start Windows.

Troubleshoot — Issues installing WSL 2 Step

9 — Enable Virtual Machine Platform

Make sure that Virtual Machine Platform is enabled on your Windows installation. You can easily check this by opening the *Turn Windows features on or off* from your Control panel.

```
C:\Users\Lenovo>docker run redis
1:C 24 Feb 2026 07:57:09.266 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
1:C 24 Feb 2026 07:57:09.266 * Redis version=7.2.4, bits=64, commit=00000000, modified=0, pid=1, just started
1:C 24 Feb 2026 07:57:09.266 # Warning: no config file specified, using the default config. In order to specify a config
 file use redis-server /path/to/redis.conf
1:M 24 Feb 2026 07:57:09.266 * monotonic clock: POSIX clock_gettime
1:M 24 Feb 2026 07:57:09.267 * Running mode=standalone, port=6379.
1:M 24 Feb 2026 07:57:09.268 * Server initialized
1:M 24 Feb 2026 07:57:09.269 * Ready to accept connections tcp
```

```
C:\Users\Lenovo>docker pull redis
Using default tag: latest
latest: Pulling from library/redis
0c8d55a45c0d: Pull complete
9a00fba7918f: Pull complete
253eb0a88de9: Pull complete
2ccab22a2cd8: Pull complete
5a312569f2e0: Pull complete
4f4fb700ef54: Pull complete
fb0cbf9357d9: Pull complete
Digest: sha256:dc123d5a424cbe44d681822c2a620f3f44bdba240b8a58a0eee0048a285532ed
Status: Downloaded newer image for redis:latest
docker.io/library/redis:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview redis
```

```
C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

```
C:\Users\Lenovo>docker ps
CONTAINER ID   IMAGE                                           COMMAND                  CREATED          STATUS          PORTS
      NAMES
1c17503a759c   170a1e90f843                                    "docker-entrypoint.s…"   5 minutes ago    Up 22 seconds   6379/
tcp   amazing_robinson
a388ed3e2ae7   docker/dev-environments-default:stable-1       "sleep infinity"         23 months ago    Up 26 seconds
      siddhesh-app-1
```

```
C:\Users\Lenovo>docker stop 1c17503a759c
1c17503a759c

C:\Users\Lenovo>docker start 1c17503a759c
1c17503a759c
```

```
C:\Users\Lenovo>docker rm 1c17503a759c
1c17503a759c
```

```
C:\Users\Lenovo>docker images
REPOSITORY                          TAG                                        IMAGE ID       CREATED         SIZE
redis                               latest                                     f0acdeb5ced3   14 hours ago    140MB
ubuntu                              latest                                     802541663949   6 months ago    78.1MB
ubuntu                              <none>                                     ca2b0f26964c   24 months ago   77.9MB
redis                               <none>                                     170a1e90f843   2 years ago     138MB
docker/dev-environments-default     stable-1                                   7eb85f44c229   3 years ago     610MB
crazymax/linguist                   7.20.0                                     dc5f9ae3be6a   3 years ago     72.2MB
docker/desktop-git-helper           5a4fca126aadcd3f6cc3a011aa991de982ae7000   efe2d67c403b   4 years ago     44.2MB
```

```
C:\Users\Lenovo>docker rmi f0acdeb5ced3
Untagged: redis:latest
Untagged: redis@sha256:dc123d5a424cbe44d681822c2a620f3f44bdba240b8a58a0eee0048a285532ed
Deleted: sha256:f0acdeb5ced3472faa8a41497498730d1b30e52ab4b8bf867b5b5ddb87720ec6
Deleted: sha256:ef393d323f971e457b565382266ff95d29e75cdcb8fa437e9bf7b73fc7313c11
Deleted: sha256:7ca2926f5c45fc35d3d02d66ad386ff0e4067527df8ca12b1b2189b185ddcfbc
Deleted: sha256:365e4cce10e83a6bb3cf41f881e4869ab7cdbaa65371796f090558e1ca2aae7e
Deleted: sha256:42a797d73e006fc25a5ca6dc274c835c061f20683294815487 5f994fabfb2d50
Deleted: sha256:fba38928ef78663422aca98187e21a3f0012559ec1a73c776b2de370967c45d3
Deleted: sha256:b599de35d6f51c157a9d80d634e4f2f4751301df3d2498f5acb35167a8d34642
Deleted: sha256:a8ff6f8cbdfd6741c10dd183560df7212db666db046768b0f05bbc3904515f03
```

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```
C:\Users\Lenovo>docker run --name ubuntu --rm -i -t ubuntu bash
root@0561ed697d99:/# exit
exit
```

```
C:\Users\Lenovo>docker run -d -p 8080:8080 ubuntu
f2de7b7b95eddddf5d839545d2e360376f90dee740084e1874c5c2c103bdcf1c
docker: Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:8080 -> 0.0.0.0:0: listen tcp 0.0
.0.0:8080: bind: Only one usage of each socket address (protocol/network address/port) is normally permitted.
```

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